Week 1

Clustering tuning

This week we ran ARCADE on the filtered MapReduce graph (569 file-level entities). We swept stop/serial thresholds `{20, 23, 30, 50, 75, 100}` for WCA (UEM and UEMNM) and LIMBO (IL), and ran ACDC once. All 19 runs finished successfully; we used the logged size balance metrics (CV, normalized entropy, singleton count, top-3 concentration) plus the A2a comparison matrix to compare partitions.

Optimized settings

WCA at threshold 100 (UEM or UEMNM):
With the low thresholds, the cluster breakdown shows almost the whole subsystem sitting in one cluster, on the order of nine tenths of all 569 files, while the rest are mostly tiny clusters. That pattern is also reflected in the evaluation metrics: normalized entropy stays very low and top-3 concentration stays very high, meaning the size distribution is heavily skewed toward a single giant cluster.
When the threshold is raised to 75 and 100, it shows a different kind of partition: the largest cluster is only on the order of a few percent of the files, and the metrics move toward more balanced sizes, higher normalized entropy, lower top-3 share, and lower coefficient of variation. Between 75 and 100, the singleton count in the log is the same, but 100 has the better balance numbers overall.
The A2a matrix backs up that this is not a small tweak: low-threshold runs are almost the same as each other, while 100 is only weakly similar to 20 (similarity on the order of ~0.08), so the tool is producing a genuinely different clustering at 100 than at the low settings.

LIMBO at 75 vs 100:
From the breakdown, LIMBO never produces the “one cluster owns almost everything” pattern that WCA shows at low thresholds; the largest group is much smaller as a fraction of the whole. Through 75, the log reports no singleton clusters; 100 adds one singleton.
On the metrics, 100 slightly wins on overall evenness (lower spread of sizes vs the mean, higher normalized entropy, less mass in the top three clusters). 75 wins on having zero singletons. A2a between 75 and 100 is fairly high (~0.78), so the two solutions are close relatives, not opposites, the log is mainly trading a bit more balance for one extra one-item cluster.

ACDC (single run):
The log gives one ACDC result: about two dozen clusters, no singletons in the reported metrics, and moderate concentration in the largest groups (top-3 around two fifths of items). Normalized entropy and coefficient of variation look reasonable compared with many other rows in the summary, not extreme skew, not trivial splitting.

Week 1

A2a tells how similar two clusterings are. ACDC is only a little similar to WCA with small thresholds (scores around 0.1). It is more similar to WCA with large thresholds and to LIMBO (scores about 0.25–0.4). So by these numbers, ACDC is closer to the WCA runs where items are split into more groups and to LIMBO, than to the WCA runs where one cluster has almost all items.

Cvg:
Every Cvg cell in the log is 1.0: all runs still cover the same set of 569 entities. So coverage does not explain why one clustering is better than another here; it only confirms everyone is clustering the same items.

summary

For WCA, only high thresholds (we pick 100) are usable on this graph; for LIMBO, 75 is the best compromise between balance and no singletons; ACDC stays the default structural comparison point.